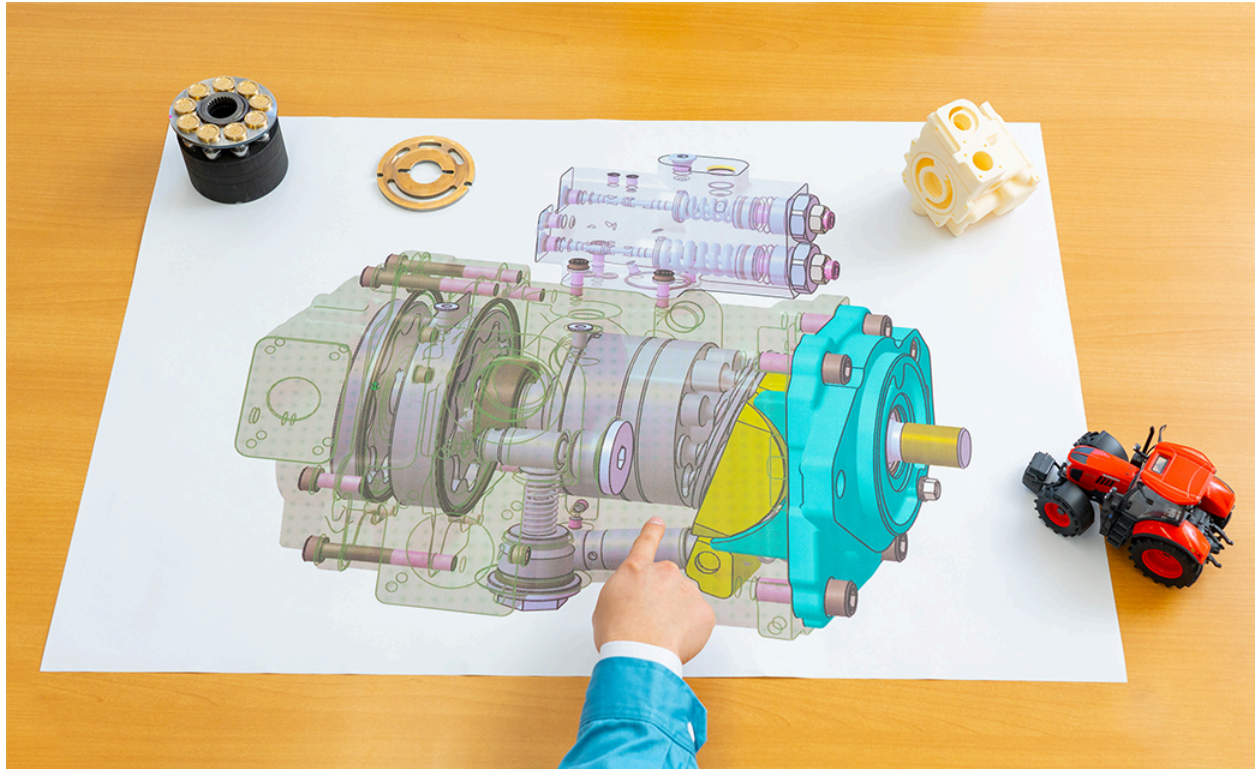




SOFTWARE REQUIREMENTS SPECIFICATION

Audrolics

Automated Calculation

Version 1.0

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Introduction

1.1 Purpose

Audrolics is a web-based system that lets field engineers build a schematic (line-and-symbol, not photorealistic) representation of a pressurized water pipe network, then uses hydraulic simulation to detect and localize anomalies (leaks, blockages, faulty readings) by comparing simulated "expected" behavior against manually entered field measurements.

1.2 Scope

In scope: schematic diagram builder, Hazen-Williams-based hydraulic simulation engine, manual measurement input, anomaly localization, landing page, authentication. Out of scope (documented as future work): ML-based condition monitoring, auto-measuring hardware, AutoCAD import, real-time collaboration, SCADA/sensor integration.

1.3 Disclaimer

This system is being built by a non-hydraulics-expert team for hydraulics practitioners (Maynilad field engineers/technicians). All calculation methods below follow the industry-standard approach used by EPANET, the most widely adopted water distribution modeling software, and should be reviewed by a hydraulics SME before production use.

1.4 Intended Users

Maynilad field engineers and technicians who need to diagram a pipe segment, enter known design values, enter field-measured values, and get a system-suggested anomaly location.

1.5 Definition of Terms

Term	Definition
Node	A point in the network (Junction, Reservoir, or Tank)
Link	A connection carrying flow between two nodes (Pipe, Pump, or Valve)

Headloss	Pressure/energy lost as water flows through a pipe due to friction
C-factor	Hazen-Williams roughness coefficient; higher = smoother pipe, less friction loss
PRV / PSV / PBV	Pressure Reducing / Sustaining / Breaker Valve
FCV / TCV / GPV	Flow Control / Throttle Control / General Purpose Valve
EPS	Extended Period Simulation — time-varying hydraulic simulation (not used in v1)
GGA	Global Gradient Algorithm — the iterative method used to solve the network equations

2. Overall Description

2.1 Product Perspective

Audrolics models a water distribution network as a graph: nodes (points with elevation/demand/storage) connected by links (pipes/pumps/valves that carry flow between nodes) — the same abstraction EPANET uses, since it's the established standard for this exact problem domain.

2.2 Tech Stack

Layer	Technology
Frontend	Next.js
Backend/API	FastAPI
Database	MongoDB
Hosting	Vercel (frontend), Render (backend)
Future ML	Modal

2.3 External Interface Requirements

Audrolics is accessed entirely through a modern web browser (latest two versions of Chrome, Edge, or Firefox) at a minimum viewport of 1280×720; no native desktop or mobile application is provided in v1. The frontend (Next.js) communicates with the backend (FastAPI) exclusively over HTTPS REST endpoints; no third-party hydraulic simulation API is used — the solver is implemented in-house per Section 4.1.

3. Feature 1: Schematic Diagram Maker

3.1 Element Library ("weights")

This is the core data model. Each element type has a fixed set of input attributes (weights: what the users enter) and computed attributes (what the simulation produces). This split matters: computed values should never be directly editable, or the anomaly detection in Feature 2 becomes meaningless.

Node Elements

Element	Symbol idea	Input attributes	Computed (by simulation)
Junction	small circle/dot	Elevation (m), Base Demand (L/s), Demand Pattern (optional, time multiplier)	Pressure Head (m), Actual Demand (L/s)
Reservoir	infinite-source symbol (e.g. hatched triangle)	Total Head / Elevation (m)	Outflow (L/s)
Tank	rectangle/cylinder	Elevation (m), Diameter (m), Min Level (m), Max Level (m), Initial Level (m)	Pressure (m), Current Volume (m ³)

Link Elements

Element	Symbol idea	Input attributes	Computed (by simulation)
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Pipe	straight line	Length (m), Diameter (mm), Roughness Coefficient — Hazen-Williams C-factor (unitless, material-dependent, e.g. ~140 for new PVC, ~100 for old cast iron), Minor Loss Coefficient (optional), Status (Open/Closed)	Flow Rate (L/s), Velocity (m/s), Headloss (m), Unit Headloss (m/km)
Pump	standard pump symbol	Pump Curve (Head vs. Flow) — see note below, or rated Power (kW), Status (On/Off), Speed setting	Flow (L/s), Head Added (m), Energy (kWh)
Valve	valve symbol, typed	Type (PRV – Pressure Reducing, PSV – Pressure Sustaining, PBV – Pressure Breaker, FCV – Flow Control, TCV – Throttle Control, GPV – General Purpose), Diameter (mm), Setting (value depends on type: pressure for PRV/PSV/PBV, flow for FCV, loss coefficient for TCV, curve data — see note below — for GPV), Status (Open/Closed/Active)	Flow (L/s), Headloss/Pressure Drop (m)
Strainer/ Filter	diamond with a dashed or straight line drawn across it	Mesh/Screen Size (mm), Minor Loss Coefficient (clean state), Status (Clean/Partially Clogged/Clogged)	Headloss (m)

Curve inputs (Pump Curve, GPV headloss curve): unlike every other Setting value, these two require multi-point data (a small set of X/Y pairs — e.g. Flow vs. Head for a pump), not a single scalar. The right panel needs a dedicated curve-editor sub-widget (a small add/remove-row table, optionally with a live-plotted preview) rather than a plain number field when either of these is selected.

Note: each valve type behaves differently in the simulation (PRV caps downstream pressure, FCV caps flow, etc.), so the right panel needs to conditionally show the correct Setting field based on which type is selected — worth flagging to whoever builds this, since it's the one part of Feature 1 with real branching logic.

3.2 UI/UX

- **Left panel:** element palette (Junction, Reservoir, Tank, Pipe, Pump, Valve), drag-or-click-to-place onto canvas

- **Right panel:** appears when an element is selected; shows its editable input attributes from the table above
- **Top panel:** general styling (line color, line thickness, symbol size, canvas zoom controls)
- **Weight-visibility toggle:** a control (e.g. checkboxes: Length | Diameter | Pressure | Flow | Elevation) that shows/hides attribute labels directly on the diagram, so the canvas isn't cluttered when the engineer only cares about one variable
- **Canvas interactions:** snap-to-connect (a pipe's endpoint locks onto a node), undo/redo, multi-select, delete, pan/zoom
- **Autosave-lite:** since full autosave is a listed recommendation for later, v1 should at minimum have an explicit **Save** button that persists the diagram (nodes, links, all attributes) to MongoDB, plus a **Load** to reopen it
- **Export:** PNG/SVG snapshot of the current diagram
- **Curve editor:** when a Pump or GPV-type Valve is selected, the right panel swaps the standard numeric Setting field for a small editable table of data point pairs, so multi-point curves can be entered without leaving the diagram view.

3.3. Input Validation Rules

Attribute	Element(s)	Rule
Elevation	Junction, Reservoir, Tank	Any real number (relative to a defined datum)
Base Demand	Junction	≥ 0 L/s
Length	Pipe	> 0 m
Diameter	Pipe, Valve	> 0 mm (typical distribution range: 12–1200 mm)
Roughness (C-factor)	Pipe	Typically 60–150 (Hazen-Williams range); reject 0 or negative
Minor Loss Coefficient	Pipe	≥ 0
Tank Min/Max/Initial Level	Tank	$\text{Min} \leq \text{Initial} \leq \text{Max}$, all ≥ 0

Pump Power	Pump	> 0 kW (if used instead of curve)
Valve Setting	Valve	> 0, unit/range depends on Type
Filter Mesh Size	Filter	> 0 mm

The system should reject a Save/Run action with a clear inline error if any attribute violates its rule, rather than allowing the simulation to run on invalid data and produce a misleading result.

4. Feature 2: Autocalculate / Anomaly Detection

4.1 Calculation Method

Headloss in pipes computed via the Hazen-Williams equation (industry default, matches EPANET):

$$h_f = \frac{K \times L \times Q^{1.852}}{C^{1.852} \times d^{4.87}}$$

Where:

K = constant based on unit system (10.67)

L = pipe length (m),

Q = flow (m³/s),

C = roughness coefficient,

D = diameter (m).

Note on simulation type: Audrolics v1 performs a steady-state simulation — a single snapshot in time, using one fixed set of demands and settings, not a time-varying (Extended Period Simulation / EPS) run. As a result, the Junction element's Demand Pattern attribute (Section 3.1) is captured in the data model as a placeholder for future EPS support but is not used in v1 calculations. Every simulation run uses each junction's Base Demand value directly, unmodified by any pattern.

Solver method: because the network can contain loops (not just a branched/tree topology), pressures and flows cannot be resolved node-by-node in sequence — they must be solved

simultaneously. Audrolics v1 uses the Global Gradient Algorithm (GGA), the same iterative method EPANET uses internally, to solve the coupled conservation-of-mass and conservation-of-energy equations across the full network per simulation run.

The system solves the network (conservation of mass at nodes, conservation of energy along loops) given the topology and input attributes, to produce expected pressure/flow at every node and link — this is the standard steady-state hydraulic simulation approach, the same underlying method EPANET uses.

4.2 Workflow

1. Engineer builds/loads the schematic with design attributes.
2. The system runs the simulation → expected pressure/flow at every node and link.
3. Actual/measured values are entered per element type: pressure readings are entered at node elements (Junction, Tank), matching where pressure gauges are physically installed in the field; flow readings are entered at link elements (Pipe), matching where flow meters are installed. A given measurement point is only valid for the attribute type that element actually produces — the UI should only offer the applicable input.
4. Engineers enter actual/measured values at multiple points across the network (the more points entered, the more precise the localization).
5. System compares expected vs. actual at each measured point, flags all points exceeding a configurable deviation threshold.
6. Segment narrowing algorithm: the system treats each pipe segment between two measured points as a candidate zone. If the mismatch pattern is consistent with a loss occurring *between* two specific points (e.g. pressure/flow makes sense up to Junction 4, but deviates from Junction 4 to Junction 7), the system narrows the anomaly to that segment rather than a single node. With more measurement points, the candidate zone shrinks further.

7. Results are shown visually on the schematic: measured points marked, and the *suspect segment* (not just a single element) highlighted — e.g. the pipe(s) between the two nearest bracketing measurement points where the mismatch first appears.

4.3 Deviation Threshold

The deviation threshold determines how large a gap between expected and measured values must be before a point is flagged as anomalous. v1 default: $\pm 10\%$ deviation on pressure and $\pm 10\%$ on flow, configurable per saved schematic (not global), adjustable by the engineer via a settings field on the diagram. The default is a starting assumption for review with a hydraulics SME — real-world sensor accuracy and normal operational variance should inform the final value before field use.

4.4 Network Validation

Before running a simulation, the system performs a pre-check and blocks the run with a specific error message (naming the offending element) if any of the following are true:

- No Reservoir or Tank (water source) exists in the network
- Any node has no connected links (orphaned element)
- Any pipe has an unconnected endpoint (dangling pipe)
- The network contains disconnected subgraphs (isolated clusters not reachable from any source)

This mirrors how EPANET itself refuses to solve an invalid network, and prevents Feature 2 from silently producing meaningless results on incomplete diagrams.

5. Feature 3: Landing Page

Static/marketing page: what Audrolics does, the problem it solves (manual leak-location calculation), how it works (diagram → simulate → compare → locate), and CTA buttons to Login/Signup.

6. Feature 4: Login & Signup

Authentication exists solely so an engineer can save their schematic diagrams and come back to them later — not for access control or team management.

- Email/password signup and login
- Each user has a personal library of saved schematics (list/load/delete past diagrams)
- No roles, no permissions tiers, no admin functions
- Session handling via standard auth (e.g. JWT or session cookie — FastAPI + Next.js supports either)

7. Non-Functional Requirements

- **Accuracy:** simulation results must match EPANET output within a reasonable tolerance for equivalent inputs (should be validated against known EPANET test networks before trusting anomaly detection in the field)
- **Performance:** simulation should resolve small-to-medium networks (order of tens of nodes) in well under a few seconds
- **Security:** passwords hashed, HTTPS in production, standard auth session handling
- **Usability:** diagram builder should be usable by engineers without prior CAD experience
- **Data integrity:** input vs. computed attributes must be clearly separated in both UI and data model so users never overwrite simulation output by mistake

8. Assumptions & Dependencies

The following assumptions inform v1 scope and should be revisited as the system matures:

- Hazen-Williams is acceptable as the default method pending confirmation from an actual Maynilad hydraulics SME
- No live sensor/SCADA integration in v1 — all "actual" values are manually typed
- Network sizes are field-segment scale, not entire city-wide distribution systems (worth confirming — this affects simulation solver choice)

9. Recommendations

The following features were considered but placed out of v1 scope; they represent natural next steps for future versions:

1. Autosave files.
2. File sharing and collaborating features that allow multiple users to edit a single design.
3. Condition Monitoring using a Machine Learning model.
4. A hardware that automatically measures the weights of each element. Hence, automatically pinpointing vulnerabilities live.
5. Automatic schematic diagram through importing existing hydraulics system design from another software like AutoCAD
6. Cater oil-based hydraulic systems.

10. Functional Requirements

The following numbered requirements consolidate the functional behavior described in Sections 3–6, for traceability during review and testing.

- FR-1: The system shall allow users to place and connect Junction, Reservoir, Tank, Pipe, Pump, Valve, and Strainer/Filter elements on a canvas.

- FR-2: The system shall allow editing of each element's input attributes via a right-side panel, including a dedicated curve-editor widget for Pump and GPV-type Valve elements.
- FR-3: The system shall validate all input attributes against the rules in Section 3.3 and block Save/Run actions on invalid data.
- FR-4: The system shall run a steady-state hydraulic simulation using the Global Gradient Algorithm upon user request, after passing the network validation checks in Section 4.4.
- FR-5: The system shall accept manually entered measured values at node elements (pressure) and link elements (flow), per Section 4.2.
- FR-6: The system shall flag measured points that deviate from simulated expected values beyond the configured threshold (Section 4.3).
- FR-7: The system shall narrow flagged deviations to a specific candidate pipe segment when multiple measurement points are available, and highlight that segment on the schematic.
- FR-8: The system shall allow authenticated users to save, load, and delete their own schematic diagrams (Section 6).
- FR-9: The system shall export the current diagram as a PNG or SVG snapshot on request.
- FR-10: The system shall present a landing page describing the product and problem it solves, with CTA buttons to Login/Signup.

11. References

Rossman, L., Woo, H., Tryby, M., Shang, F. — EPANET 2.2 User Manual, U.S. Environmental Protection Agency.

Romero Ben, L., Alves, D., Blesa Izquierdo, J., Cembrano Gennari, G., Puig Cayuela, V., & Duviella, E. (2023). Leak detection and localization in water distribution networks: Review and perspective. *Annual Reviews in Control*, 55, 392–419.

Additional leak-detection and anomaly-localization literature, as selected for the project's related-work review (graph-based localization, ML-based condition monitoring surveys). ISO/ASCE reference tables for Hazen-Williams C-factor by pipe material, as applicable.

Appendix A: Units & Conventions

Attributes intentionally mix meters and millimeters (elevation/length vs. diameter), matching common field practice. Implementers should not silently normalize all values to one unit, as this would break the formulas in Section 4.1.

Quantity	Unit
Length, Elevation	meters (m)
Diameter	millimeters (mm)
Flow	liters/second (L/s)
Pressure/Head	meters of head (m); bar shown via UI toggle
Power	kilowatts (kW)